

Vibrational Natural Optical Activity in Periodic Systems

A first-principles spatial-dispersion framework

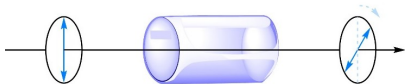
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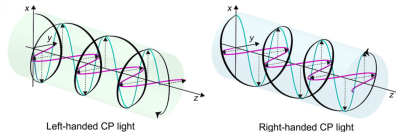
May 2026

Natural Optical Activity

Empirical manifestations (e.g. in **chiral materials**)



Optical rotation



Circular dichroism

- **Natural optical activity tensor** = first-order spatial dispersion of the dielectric tensor

$$\varepsilon_{ab}(\mathbf{q}, \omega) = \varepsilon_{ab}(\omega) + iq_{\gamma} \eta_{ab,\gamma}(\omega) + \dots$$

- Restrict to **infrared** spectrum: **vibrational natural optical activity (VOA)** and **vibrational circular dichroism (VCD)**

VCD: Motivation and Established Theory

Interest in VCD in last 50 years...

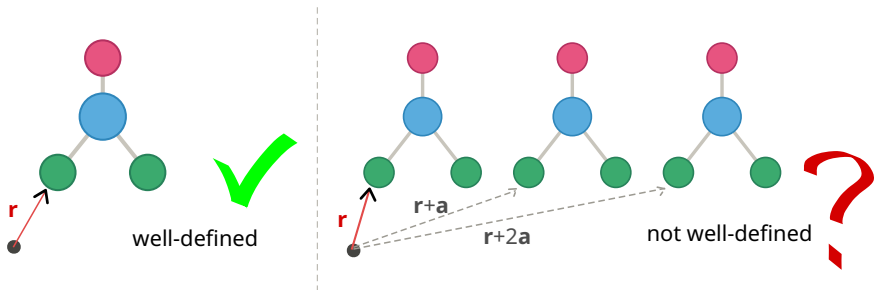
- Practical application: determine absolute configuration of chiral materials
- Key tool in pharmaceutical industry to distinguish enantiomers with different biological effects
- Theory essential for interpretation of VCD measurements

Established theories for molecules:

- Magnetic Field Perturbation (MFP) and Nuclear Velocity Perturbation (NVP)
[Stephens 1985, Nafie et al. 1983, Buckingham et al. 1987]
- Implemented in DFT codes
- Giving reliable results

Limitations of existing VOA theories

- Only apply to **finite** systems
- They **cannot be directly applied to periodic systems** – they rely on the **explicit position operator**, which is **incompatible with crystal translational invariance**



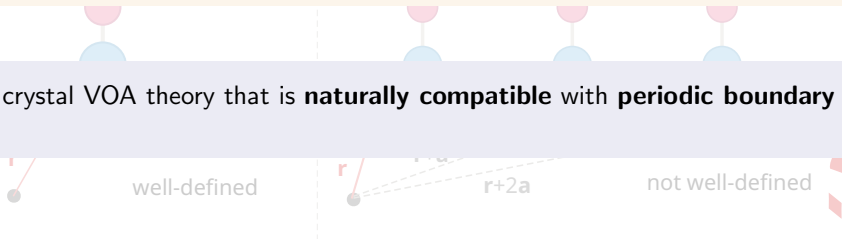
- VCD spectra of crystals look *qualitatively different* from molecular ones
- Supercell approach [Jähnigen 2023] is limited by system size

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⚠ No first-principle theoretical framework for crystal VOA with periodic boundary conditions

🎯 AIM: crystal VOA theory that is **naturally compatible** with **periodic boundary conditions**



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💡 **SPATIAL-DISPERSION based approach for VOA**

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Optical conductivity tensor

$$\sigma_{ab}(\mathbf{q}, \omega) = -\frac{i\omega}{4\pi} (\varepsilon_{ab}(\mathbf{q}, \omega) - \delta_{ab}) \quad \rightarrow \quad \text{current density response to an external electric field}$$

- infrared absorption dominated by vibrational transitions

$$\Delta\varepsilon_{ab}(\mathbf{q}, \omega) = \frac{4\pi}{\Omega} [Z_a(\mathbf{q}) \cdot U^{-1}(\mathbf{q}, \omega) \cdot Z_b^*(\mathbf{q})], \quad U^{-1}(\mathbf{q}, \omega) = \sum_{\nu} \frac{|\nu\mathbf{q}\rangle\langle\nu\mathbf{q}|}{M(\omega_{\mathbf{q}\nu}^2 - \omega^2)}$$

- static approximation for polarization response $Z(\mathbf{q})$ and force constant matrix
- consider Hermitian part $\rightarrow$ dissipative (VCD)

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- static approximation for polarization response $Z(\mathbf{q})$ and force constant matrix
- consider Hermitian part $\rightarrow$ dissipative (VCD)

$$\Delta\sigma_{ab}^H(\mathbf{q}, \omega) = \frac{2\pi}{\Omega} \sum_{\nu} D_{\alpha\beta}^{(\nu)}(\mathbf{q}) [L^{\eta}(\omega_{\mathbf{q}\nu} - \omega) + L^{\eta}(\omega_{\mathbf{q}\nu} + \omega)]$$

Momentum-dependent oscillator strength $D_{ab}^{(\nu)}(\mathbf{q}) = Z_{a,\mathbf{q}\nu} Z_{b,\mathbf{q}\nu}^*$, $Z_{a,\mathbf{q}\nu} = \frac{1}{\sqrt{M_{\alpha}}} \langle Z_{a,k\alpha}(\mathbf{q}) | \nu\mathbf{q} \rangle$

Long-wavelength limit

$$\Delta\sigma_{ab}^H(\mathbf{q}, \omega) = \Delta\sigma_{ab}^H(\omega) + \sum_{\gamma} q_{\gamma} \Delta\sigma_{ab,\gamma}^H(\omega) + \dots$$

- at first-order:

$$\Delta\sigma_{ab,\gamma}^H(\omega) = \frac{2\pi}{\Omega} \sum_{\nu} \left\{ \left. \frac{\partial D_{ab}^{(\nu)}(\mathbf{q})}{\partial q_{\gamma}} \right|_{\mathbf{q}=0} [L^{\eta}(\omega_{\mathbf{q}\nu} - \omega) + L^{\eta}(\omega_{\mathbf{q}\nu} + \omega)] \right. \\ \left. + D_{ab}^{(\nu)}(\mathbf{q}) \frac{\partial [L^{\eta}(\omega_{\mathbf{q}\nu} - \omega) + L^{\eta}(\omega_{\mathbf{q}\nu} + \omega)]}{\partial q_{\gamma}} \right|_{\mathbf{q}=0} \Bigg\}$$

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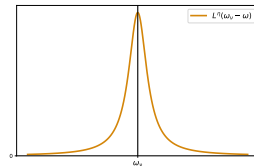
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Derivative of the Oscillator Strength

$$\Delta\sigma_{ab,\gamma}^{H1}(\omega) = \frac{2\pi}{\Omega} \sum_{\nu} D_{ab,\gamma}^{(\nu)} [L^{\eta}(\omega_{\nu} - \omega) + L^{\eta}(\omega_{\nu} + \omega)]$$

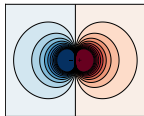


- Dispersion of the **polarization response**: $|Z(\mathbf{q})\rangle = |Z\rangle - iq_{\gamma}|Z^{\gamma}\rangle$

Z

dynamical dipoles

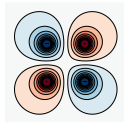
Born effective charge



Z^{γ}

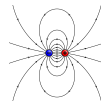
dynamical quadrupoles

symmetric



dynamical orbital magnetic moment $\mathcal{M}$

antisymmetric



- Dispersion of the **phonon eigenvector**: $D(\mathbf{q}) = D - iq_{\gamma}D^{\gamma}$, $|\partial_{\gamma}\nu\rangle = \sum_{\omega_{\mu} \neq \omega_{\nu}} \frac{\langle \mu | D^{\gamma} | \nu \rangle}{\omega_{\mu}^2 - \omega_{\nu}^2} |\mu\rangle$

Derivative of the Oscillator Strength

$$D_{ab,\gamma}^{(\nu)} = i \left[Z_{a\nu}^{\gamma} Z_{b\nu} - \sum_{\omega_{\mu} \neq \omega_{\nu}} Z_{a\mu} \frac{\langle \mu | D^{\gamma} | \nu \rangle}{\omega_{\mu}^2 - \omega_{\nu}^2} Z_{b\nu} \right] + h.c.$$

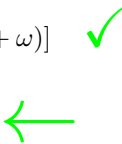
- E1-E2 and E1-M1 couplings
- Nontrivial mixing of phonon modes at first order in q

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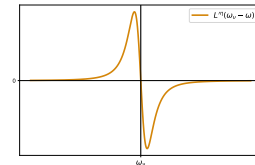
- E1-E2 and E1-M1 couplings
- Nontrivial mixing of phonon modes at first order in q
- Recovers the established quantum-chemistry formula for VCD
- $D_{ab,\gamma}^{\nu}$ is gauge invariant, individual terms are not (gauge arbitrariness of phonon eigenvectors)

Derivative of the Lorentzian

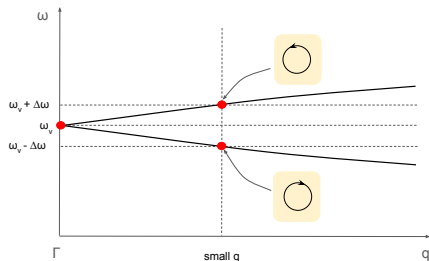
$$\Delta\sigma_{ab,\gamma}^H(\omega) = \frac{2\pi}{\Omega} \sum_{\nu} \left\{ \left. \frac{\partial D_{ab}^{(\nu)}(\mathbf{q})}{\partial q_{\gamma}} \right|_{\mathbf{q}=0} [L^{\eta}(\omega_{\mathbf{q}\nu} - \omega) + L^{\eta}(\omega_{\mathbf{q}\nu} + \omega)] \right. \\ \left. + D_{ab}^{(\nu)}(\mathbf{q}) \frac{\partial [L^{\eta}(\omega_{\mathbf{q}\nu} - \omega) + L^{\eta}(\omega_{\mathbf{q}\nu} + \omega)]}{\partial q_{\gamma}} \right|_{\mathbf{q}=0} \Bigg\}$$


Derivative of the Lorentzian

$$\Delta\sigma_{ab,\gamma}^{H2}(\omega) = \frac{2\pi}{\Omega} \sum_{\nu} D_{\alpha\beta}^{(\nu)} [L'^{\eta}(\omega_{\nu} - \omega) + L'^{\eta}(\omega_{\nu} + \omega)] \left. \frac{\partial\omega_{\nu}}{\partial q_{\gamma}} \right|_{\mathbf{q}=0}$$



- Non-vanishing *only* for phonons with linear momentum-dispersion at $\Gamma \rightarrow$ E modes



No dispersion of degenerate phonons in finite systems $\rightarrow$ new effect arising from periodicity

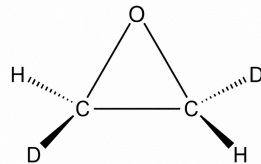
- ✓ Building blocks (Z, Z^γ, D^γ) **well-defined under periodic boundary conditions**
- ✓ Available in the *longwave* driver of **ABINIT**
[Royo et al. 2019]
- ⚠ Implementation of travelling pseudopotentials correction for Z^γ
[Stengel et al. 2025]

$$\text{Piezoelectric sum rule: } - \sum_k Z_{\alpha,k\beta}^\gamma = \Omega \bar{e}_{\alpha,\beta\gamma}$$

Molecular validation: R-d₂-Oxirane

- Benchmark chiral molecule, widely studied in VCD literature
- Compare rotational strengths with **literature**
- Ditler et al. 2022**: CP2K package implementation of both NVP and MFP theories

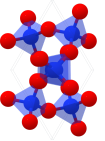
Comparison	RMSD
NVP - MFP	20%
our results - NVP	10%
our results - MFP	17%



✓ Our spatial-dispersion formalism is **quantitatively reliable**

Test Crystal: α -Quartz

- Structure: trigonal chiral space group **P3₁21** with screw axis along z (optical axis)



Mode	Freq. [cm ⁻¹]	Character
4,5	129	E
6	220	A ₁
7,8	258	E
9	342	A ₁
10	348	A ₂
11,12	379	E
13,14	441	E
15	457	A ₁
16	487	A ₂
17,18	689	E
19	767	A ₂
20,21	790	E
22,23	1057	E
24	1067	A ₂
25	1073	A ₁
26,27	1143	E

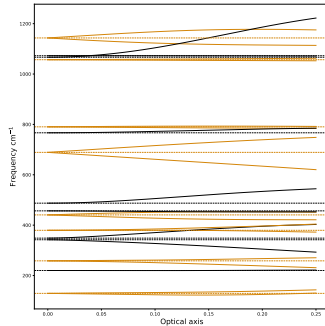


Figure: Phonon bands of α -quartz along z

- Degenerate E modes **split at finite q_z** $\rightarrow$ “crystal specific” contribution to VOA

VCD Spectrum of α -Quartz

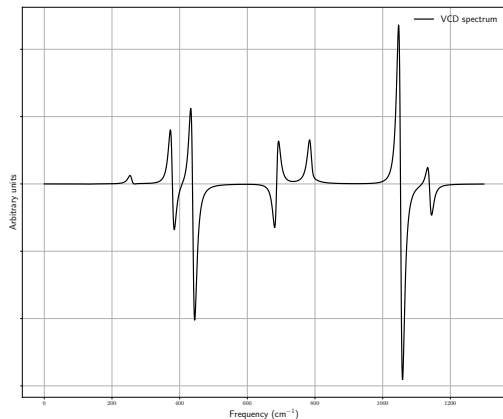
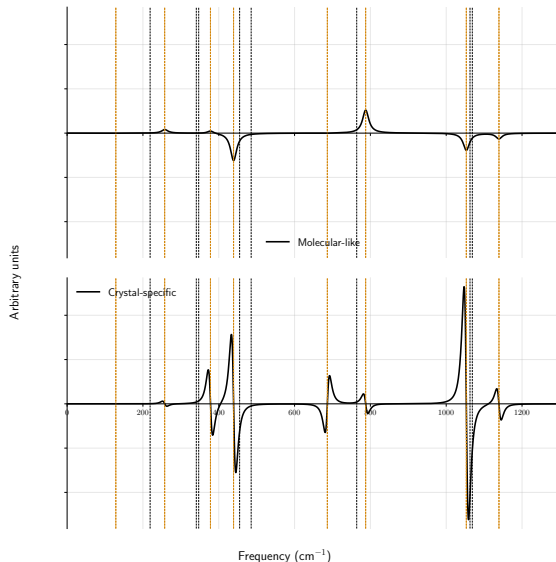


Figure: Total VCD spectrum of α -quartz (with imaginary broadening of 5 cm^{-1})

- Prediction of VOA of a chiral crystal with *ab initio* DFT calculation enforcing periodicity
- Next: decompose into its mode-projected contributions

Molecular-like vs. Crystal-Specific Contributions



- “Crystal-specific” term is dominant
⇒ periodicity determinant in VOA

Summary and Outlook

Key results of our spatial-dispersion approach to VOA:

- ✓ **PBC compatibility**
- ✓ Reproduce **standard results** in molecular limit
- ✓ Building blocks **accessible** in the ABINIT *longwave* driver

Next steps:

- Direct comparison with solid-state VCD experiments

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Key results of our spatial-dispersion approach to VOA:

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Thank you!

Questions?

Quantum chemistry theories

In quantum chemistry, in terms of microscopic current response to displacement of isolated atom $\frac{\partial \mathbf{J}(\mathbf{r})}{\partial \dot{\mathbf{R}}_{k\alpha}}$

- $Z \rightarrow \int d^3r \frac{\partial \mathbf{J}(\mathbf{r})}{\partial \dot{\mathbf{R}}_{k\alpha}}$: Atomic Polar Tensor
- $\mathcal{M} \rightarrow \frac{1}{2} \int d^3r (\mathbf{r} - \mathbf{R}_k) \times \frac{\partial \mathbf{J}(\mathbf{r})}{\partial \dot{\mathbf{R}}_{k\alpha}}$: **intrinsic** Atomic Axial Tensor

Rotational strength: $R_\nu = \tilde{\mathcal{M}}^{(\nu)} \cdot Z^{(\nu)}, \quad \tilde{\mathcal{M}}_{a,k\alpha} = \frac{1}{2}(\mathbf{R}_k \times \mathbf{Z}_{k\alpha})_a + \mathcal{M}_{a,k\alpha}$



Only magnetic dipole - electric dipole interactions appear: not electric quadrupole - electric dipole

- For molecules in solution, one needs to consider the orientationally averaged VCD signal = trace of the gyration tensor $\rightarrow$ electric quadrupole cancels out

$$g_{ab}(\omega) = \frac{1}{2} \epsilon_{\delta a} \eta_{\gamma \delta, b}(\omega)$$

Molecular limit

- Recovers the established quantum-chemistry formula for the rotational strength

$$D_{ab,\gamma}^{(\nu)} = i \left[Z_{a\nu}^{\gamma} Z_{b\nu} - \sum_{\omega_{\mu} \neq \omega_{\nu}} Z_{a\mu} \frac{\langle \mu | D^{\gamma} | \nu \rangle}{\omega_{\mu}^2 - \omega_{\nu}^2} Z_{b\nu} \right] + h.c.$$

- In molecules, the spatial dispersion of D has a purely *geometric* character

$$D_{\kappa\alpha,\kappa'\beta}^{\gamma} = D_{\kappa\alpha,\kappa'\beta} (R_{\kappa\gamma} - R_{\kappa'\gamma})$$

- The (origin-dependent) atomic position is reabsorbed in the first term via

$$\tilde{Z}_{a,\kappa\alpha}^{\gamma} = Z_{a,\kappa\alpha}^{\gamma} + Z_{a,\kappa\alpha} R_{\kappa\gamma}$$

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Generalized crystals case:

- Spatial dispersion of D reflects nontrivial mixing of phonon modes at first order in q
- $D_{ab,\gamma}^{\nu}$ is gauge invariant, individual terms are not (gauge arbitrariness of phonon eigenvectors)